

AI for All Minds: Neurodiversity in Higher Education — A Research Brief for Josh Penzell's Keynote

TL;DR

- **Generative AI is functioning as a de facto accommodation for neurodivergent (ND) college students faster than any institution is formally recognizing it.** Empirical 2024–2025 studies show ND students use AI primarily to externalize executive function, translate between neurotypes, and compensate for under-resourced Disability Services Offices — often without ever disclosing a diagnosis.
- **The same systems that help are simultaneously harming ND students through AI-detection false positives, ableist training data, and pressure to "humanize" (i.e., mask) authentic ND communication styles.** Institutional policy, not the technology itself, is the lever that decides whether AI amplifies inclusion or amplifies exclusion.
- **Josh's "AI amplifies rehearsal, not intelligence" thesis maps cleanly onto what the empirical literature is finding.** AI's measurable benefit to ND students is overwhelmingly in scaffolding, drafting, iteration, social scripting, and task initiation — the rehearsal moves — not in cognitive augmentation per se.

Key Findings

1. Prevalence and the scale of the issue

- Nancy Doyle's widely cited estimate (*British Medical Bulletin*, 2020) puts **15–20% of the global population as neurodivergent**, encompassing ADHD, autism, dyslexia, dyspraxia, dyscalculia and related profiles. (Neurodiversity) (Sage Journals)
- YouGov (2024) found **19% of Americans self-identify as neurodivergent**; 47% know someone who does. (Yougov)
- In U.S. higher ed specifically, the most recent NCES data (2019–20, *Digest of Education Statistics* Table 311.10) report **21% of undergraduates and 11% of postbaccalaureate students** as having a disability. A Purdue institutional study found ~36% of incoming students endorsed at least one neurodivergent identity category despite only 6–7% being registered with the Disability Resource Center — a **roughly 5x disclosure gap**. (NCES + 2)
- UC Davis Academic Senate report: graduation rate for ND students is **58% vs. 79% for non-disabled students**; "more than half of autistic young adults do not pursue a college degree." (University of California Aca...) (University of California Aca...)
- Duke's DUCkI sample: 10.2% of students identified as neurodivergent; **64.7% of ND students also identified as LGBTQIA+** (vs. 17.3% of non-ND students) — a major intersectional fact for keynote framing. (Duke)

2. The single most important study for the keynote

Atcheson, Khan, Siemann, Jain & Karahalios (CHI 2025) — *"I'd Never Actually Realized How Big An Impact It Had Until Now": Perspectives of University Students with Disabilities on Generative Artificial Intelligence*. (ACM CHI '25, Article 42; 62 survey + 21 interview participants across two large public U.S. universities.) [Journal of the Association f...](#)

Headline findings:

- Median student used **2 GenAI tools**; ChatGPT (n=55) dominated. [uiuc](#)
- **44 of 62** said GenAI had at least a slight disability-related impact.
- **25 students agreed** (vs. 17 who disagreed) that GenAI supported them *"in important ways that the university does not."*
- Of 39 students with formal accommodations, **13 said GenAI had changed how they use at least one accommodation.**

Quotes Josh can use directly:

- **P43:** *"I have a written expression disorder and struggle to communicate adequately in writing. AI gives me a draft to work with and reduces the immense cognitive load of writing."* [uiuc](#)
- **P31:** *"I have used ChatGPT to help me get accommodations through the university, and also to help me communicate more directly and assertively with professors about my accommodations and needs."* [uiuc](#)
- **P10:** *"the bureaucracy of it all, versus with AI, it's just right there. There's no limitations, you don't have to go through paperwork... And also AI doesn't require documentation of a disability."* [uiuc](#)

That last quote is the political dynamite of the talk: **AI is a workaround for medical-model gatekeeping.**

3. The companion European study

Pierrès, Darvishy & Christen (2024/2025), *Education and Information Technologies* 30:8923–8952 — 33 disabled students interviewed at Swiss/French/German universities. Found ChatGPT used as "an assistant in teaching, writing, reading and research, or self-organization." Key concerns: accuracy/hallucination, **financial barriers (paid tiers exclude disabled students who "often face financial constraints")**, privacy of disability disclosure, ableist outputs, and digital multitasking risk for ADHD users. [ACM Digital Library + 6](#)

4. The UK survey (Zhao, Cox & Chen 2025, *Internet and Higher Education*)

124 disabled students at a UK university: [White Rose Research Online](#)

- **77% used GenAI; 91% used only free tiers** — a stark digital divide. (whiterose)
- Conditions: ADHD/ND **34.7%**, dyslexia/dyspraxia **29%**, autism/social communication **21.8%**, mental health **14.5%** (overlapping). (White Rose Research Onlin...)
- Top tasks: summarizing readings, overcoming mental blocks, brainstorming, rewriting phrases, structuring ideas. (whiterose)
- Verbatim autistic student: *"It impacts in more ways than I can list. My entire process of how I do my academic writing will be very different to that of someone who is not autistic because our internal processes are different."* (White Rose Research Online) (whiterose)

5. The Nottingham Trent study (He, Chen & Fletcher 2024)

54 disabled students at Nottingham Trent University; **59% had used GenAI to explain concepts**; only 9% used it as a "private tutor"; **50% reported being more worried about plagiarism accusations since GenAI emerged** — a direct empirical link to the detector-bias problem below.

(Nottingham Trent Universit...)

6. The autistic ChatGPT meta-study

Ma et al. (2026), **"I use ChatGPT to humanize my words": Affordances and Risks of ChatGPT to Autistic Users** (arXiv:2601.17946) — inductive thematic analysis of **3,984 social media posts** from self-identified autistic users. Found four affordances: (1) offloading executive tasks, (2) emotional regulation, (3) translation between ND/NT communication, (4) validating autistic identity via "algorithmic mirroring." But three risks: **(a) reinforcing delusional thinking; (b) "replacing authentic identity with automated masking"; (c) triggering conflict between ChatGPT's "functional unity" and the autistic sense of justice.** This is the central tension for Josh's talk — **the "humanize my words" framing is itself an indictment:** AI is being used by autistic adults to launder their voice into neurotypical-acceptable form. That is masking-as-a-service. (arXiv + 5)

7. The AI-detector bias crisis

- Liang et al., **"GPT detectors are biased against non-native English writers"** (*Patterns* Vol. 4, July 2023; arXiv:2304.02819): on 91 TOEFL essays, seven commercial detectors averaged a **61.22% false-positive rate**; 97.80% of the essays were flagged as AI-generated by at least one detector. (Fast.ai + 2)
- HEPI Policy Note 61, **"Student Generative AI Survey 2025"** (Josh Freeman, Feb 2025; n=1,041 UK undergrads via Savanta): **92% of students now use AI in some form (up from 66% in 2024); 88% use it specifically for assessments (up from 53%); 18% have included AI-generated text directly in their work**, though the report notes "little evidence here that AI tools are being misused to cheat." (Smart Thinking + 2)
- **Bloomberg Businessweek (Davalos & Yin, Oct 2024):** profiled an autistic student flagged at 100% AI use by a detector for writing "in the same formulaic manner as AI"; she was

vindicated but warned not to be flagged again. (AI Detector Pro)

- **The Rua Mae Williams "I'm just Autistic" exchange** (Purdue, July 2023) — a faculty member's research-inquiry email was rejected by a recipient who wrote: *"The AI design of your email is clever, but significantly lacks warmth. Would it be possible to be contacted by a human being moving forward instead of AI?"* Williams' reply: **"It's not an AI. I'm just Autistic."** Williams told the *Purdue Exponent*: *"There's really not something that disabled students can do to protect themselves. We really need to be focusing on faculty pedagogy and that they're not putting students in a position where they could be flagged falsely for cheating."* Williams' forthcoming Palgrave book is titled *Disabling Intelligences: Legacies of Eugenics and How We Are Wrong About AI* — Josh should cite Williams as the leading critical-disability voice on this issue. (Medium) (Purdue Exponent)

8. UDL × Generative AI is now the dominant teaching frame

- CAST (architects of UDL) released **UDL Guidelines v3.0 (2024)** which explicitly integrate AI considerations.
- **Mallary (2025), *New Directions for Adult and Continuing Education*** provides the most cited adult-ed framework on UDL+GenAI integration, covering AI-generated rubrics, adaptive feedback, multilingual support; flags algorithmic bias and data privacy as primary risks. (Wiley Online Library)
- California Virtual Campus, the University of British Columbia CTLT, and USC Upstate now run faculty workshops on "Generative AI Meets UDL."
- **The systematic review by Liao (ASIS&T 2025)** identifies 21 studies (2022–2025) on GenAI for ND students in HE; **Dumitru et al. (2026, *Sage Journals*)** reviewed 27 papers (2020–2024) and named five themes: personalized learning, AT benefits, adoption challenges, institutional barriers, best practices. (Journal of the Association f...)

9. Specific tools the ND student population is using

- **Goblin Tools** (Bram De Buyser): free; "Magic ToDo" breaks tasks down via LLM; "Formalizer" rewrites brain-dump into professional tone (especially valued for emails); "Estimator" addresses time blindness. Widely cited by autistic/ADHD users as the executive-function tool. (Focushack + 2)
- **Otter.ai & Glean (now Genio)** are standard accommodation tools at Berkeley DSP, Harvard, UMBC, UCCS, and UCI — provided as free licenses through Disability Services. Glean takes an "active curation" approach (notes synced to slides + audio); Otter takes a "real-time transcription" approach. (AFFINE)
- **Speechify, Read&Write (Everway), Kurzweil 3000, Snap&Read** dominate dyslexia support; Snap&Read's "text leveling" feature simplifies college-level prose while preserving meaning. (CastReader)

- **NotebookLM** (Google) emerged in 2025 as a major study tool — Beacon College's learning specialists specifically discuss it. (Class Tech Tips)
- **Custom GPTs** are being created by faculty to act as 24/7 tutors aligned to course materials.

10. Disability Services Offices in transition

- **PLOS One (2025)**, "**Disability services in higher education: statistical disparities and the potential role of AI**": documents 9.47% accommodation provision at two-year vs. 28.40% at four-year institutions; staff-to-student ratios of 1:200 (two-year) vs. 1:75 (four-year). (PubMed Central)
- The **National Deaf Center (2025)** is the most explicit advocate so far for treating AI as a formal accommodation, citing the 2025 NASPA Disability Knowledge Community meeting. (National Deaf Center)
- **AHEAD** is running "AI for Inclusion" webinars but has not yet issued formal guidance.
- **Teachers College, Columbia (2023)** offered the prototype syllabus language: *"Students with disabilities are eligible for reasonable accommodations to permit them equal access... If you are a student registered with [DSO] with a generative AI accommodation, please speak with me directly about your needs."*
- **Inside Higher Ed (Nov 2025)**: *"If a tool helps level the playing field for students with disabilities or other disadvantages, ignoring it is not neutrality—it's negligence."* (Inside Higher Ed)

11. Aphantasia and AI (Josh's personal hook)

- **Hollis Robbins (Aphantasia Network)**: people with aphantasia (~4% of the population) have always demonstrated that you can read and learn without mental imagery; she argues the rise of GenAI may finally let "visualization pedagogy" be retired in favor of code-pattern reading, vindicating the aphantasic literacy experience. Verbatim: *"I and hundreds of millions of others around the world with no 'mind's eye' have learned to read without visualization."* (Aphantasia Network)
- First-person testimony (**DEV.to**, "I Have Aphantasia and Can't Visualize"): *"For the first time in my life, I could describe an image in words and see it materialize. Not in my mind (that's still black), but on a screen, in seconds. The gap between concept and visual output — the gap that aphantasia creates — was suddenly bridgeable."* This is precisely Josh's keynote moment: **AI as externalized imagination.** (DEV Community)
- Emerging neuroscience: 7-Tesla fMRI work shows aphantasic brains generate imagery but lack the feedback loop to consciousness; researchers are training models to decode fMRI imagery via Stable Diffusion. Research-stage, not deployed — Josh should flag accordingly. (Kerson AI Solutions)

12. Programs and institutional examples

- **Landmark College (Putney, VT)** — the only U.S. college exclusively for ND students. The Landmark College Institute for Research and Training (LCIRT) has run CAST webinars on AI for ND students, led by Michelle Deal, Director of Learning Technologies R&D. Per *Government Technology*: "I think AI should be an accommodation that can be used with students." The new **AIE-STEMPLOS program** at Landmark, published in the *Journal of Science Education for Students with Disabilities* (March 2026), is co-authored by four ND undergraduates and uses UDL + strengths-based mentoring. (Landmark College + 5)
- **Beacon College (Leesburg, FL)** — first U.S. accredited bachelor's-degree institution dedicated to students with LD/ADHD. 2025 named to **Forbes Accessibility 100**. Learning specialists Sam Owens and Shauna Misir are publicly discussing AI integration (NotebookLM, Goblin Tools, Copilot) and have introduced an "Intro to College Academics" course explicitly covering AI as assistive technology. (Beacon College + 2)
- **Special Olympics Global Center for Inclusion in Education** (Tim Shriver, Mary Davis) — partnered with Stratalys Research for a survey of **500 U.S. parents of children with IDD and 200 K-12 teachers (fielded June 3-10, 2024)**. Headline: **64% of educators and 77% of parents** view AI as a powerful mechanism for inclusion, but **only 35% of educators believe developers currently account for the needs and priorities of students with IDD**. Shriver: failure to listen will "result in three percent of the population being locked out of the most revolutionary technology since the advent of the personal computer." Special Olympics International CIO **Nathan Cook** describes the mantra "**Nothing about us, without us**" as the guide for their AI work — this aligns directly with Josh's Inclusion Academy work. (Education Week + 10)
- **Think College / Inclusive Higher Education Network** (UMass Boston; Meg Grigal & Clare Papay) — the national TA center funded by the U.S. Department of Education for 259 inclusive postsecondary programs for students with intellectual disability. Critical context: Think College has not yet released a formal AI position paper, leaving a gap Josh could speak into. (Thinkhighered + 2)
- **UC system** — UC Davis Aggie Neurodiversity Committee has formal recommendations endorsed by UCADDE; UC Berkeley DSP provides free Glean/Otter licenses; the UC Council on Affirmative Action, Diversity and Equity has flagged ND as the lowest-graduation-rate disability group. (Berkeley Disabled Student...)
- **Cornell, Curry College PAL, Marshall University West Hall, Drexel** all run autism-support programs; **Curry College** is particularly known for its longstanding "PAL" model.
- **International**: UK's **Jisc** runs a National Centre for AI in Tertiary Education (2024-2025 pilots with Jamworks); Australia's National AI in Schools Framework explicitly references disability access (Leslie Loble's policy work); HEQCO Ontario's November 2024 report on AI for accessibility is the most substantive Canadian source. (Calstate)

- **Rua Mae Williams (Purdue)** — leading scholar; CoLiberation Lab; book *Disabling Intelligences: Legacies of Eugenics and How We Are Wrong About AI* (Palgrave, forthcoming). [Purdue Polytechnic Institute](#)
- **Devon Price (Loyola Chicago)** — *Unmasking Autism* (2022): warns that masking is "rewarded and facilitated by neurotypical people" because it "renders us compliant and quiet." Direct relevance to AI: tools that "humanize" autistic writing operationalize masking at scale. [Goodreads](#) [Goodreads](#)
- **Hamilton & Petty (2024)**, *Frontiers in Psychology*, "Compassionate pedagogy for neurodiversity in higher education": argues current HE contexts present "a conditional view of an acceptable student, i.e., a student is acceptable only when fitting to a neurotypical standard." [nih](#)
- **Nathan Cook (Special Olympics, on *We Are For Good* podcast)**: "AI has an ableism problem." When Special Olympics analyzed how large language models describe people with disabilities, outputs were systematically deficit-framed. [We Are For Good](#)

14. Harms / where to push back

1. **AI-detector false positives** disproportionately flag autistic and ESL students (Liang et al.'s 61.22% false-positive rate on TOEFL essays). [Fast.ai](#) [arxiv](#)
2. **Atrophy of skills** — *Frontiers in AI* (Sweden, 2024): students with low EF derive the greatest benefit from EF-improving activities; over-reliance on GenAI may displace the practice that builds EF. [Frontiers](#)
3. **Subscription divide** — 91% of disabled UK students in Zhao et al. could only access free tiers; the best models are paywalled. [whiterose](#)
4. **Surveillance/Plagiarism panic** — Nottingham Trent: 50% of disabled students *more* worried about plagiarism accusations since GenAI. [Phys.org](#)
5. **Identity erasure** — Ma et al.: AI as "automated masking" replaces authentic autistic voice. [arxiv](#)
6. **Algorithmic ableism in training data** — Special Olympics documented LLMs describing disabled people in deficit-framed ways.
7. **Bias against atypical writing patterns** — autistic writing's repetition and directness gets flagged. [Purdue Exponent](#) [Medium](#)
8. **Hallucination** — particularly harmful for students who already struggle to challenge authority/text.

Details — Recommended Keynote Architecture

Opening hook (Josh's personal aphantasia story → keynote thesis):

"For 39 years I couldn't see a red apple when I closed my eyes. Then ChatGPT could draw it. That gap — between concept and image, between thought and shared artifact — isn't intelligence. It's rehearsal. And rehearsal is what every neurodivergent mind has been doing in silence since the day we started school."

Section 1 — The Data Gap: 15–20% of the population is ND; only ~7% of college students are registered with disability services; AI adoption rates among UK students are 92% overall and 88% for assessments. AI is already the universal accommodation that the institution never granted. (Smart Thinking)

Section 2 — What ND students actually do with it (anchor on Atcheson et al. CHI 2025 quotes): self-advocacy drafting, bypassing bureaucracy, executive function offload, social translation.

Section 3 — The harm: Rua Williams' email, the autistic student flagged at 100%, masking-as-a-service ("I use ChatGPT to humanize my words").

Section 4 — TheaterThink framing: rehearsal-room principles applied to AI work. AI as the cold read, the table read, the run-through — not the performance. The student is still the actor; AI is the dramaturg, the line-feeder, the prompter in the wing.

Section 5 — The faculty mandate: UDL v3.0 + permissive AI policy + ND-aware syllabus language. Cite Teachers College Columbia's prototype language.

Section 6 — The Inclusion Academy call to action: Nothing about us, without us. Special Olympics' 35% gap. Build with ND students, not for them.

Recommendations

For the keynote audience (higher ed faculty, instructional designers, DSO leaders), Josh should make three concrete asks, each with a clear benchmark:

1. **Rewrite your syllabus before fall 2026 to include disability-affirming AI language.** Benchmark: adopt or adapt the Teachers College Columbia model statement. Trigger to revise downward: campus-wide AI ban. Trigger to expand: any formal OCR or ADA guidance on AI as accommodation.
2. **Stop using AI-detection software as an evidentiary basis for academic-integrity charges.** Benchmark: Liang et al.'s 61.22% false-positive rate on non-native English samples. If your detector has not been independently audited for false-positive rates across ND and ESL populations and shown to be below ~2%, you cannot ethically use it against autistic, ESL, or formulaic-writing students. No detector currently meets that bar. (arxiv)
3. **Fund free-tier-equivalent AI access through Disability Services.** Benchmark: 91% of disabled students in Zhao et al. (UK) were locked into free tiers. UC Berkeley's free Glean/Otter licensing model is the floor; the ceiling is institutional Claude/ChatGPT/Copilot

Enterprise seats routed through DSO. Trigger to expand: any documented inequity in AI tool use between registered-disability and non-disability student populations. whiterose

A subtler institutional recommendation: **co-design with ND students**. Atcheson et al.'s direct recommendation is "involve disabled students as co-designers." Special Olympics' Nathan Cook: "nothing about us, without us." Josh's Inclusion Academy work is the operational model the audience can adopt. We Are For Good

Caveats

- The empirical literature is **less than 30 months old in most cases**; almost no longitudinal outcome data (retention, GPA, completion) yet exists for AI-assisted ND students. Studies cited are small (33, 54, 62, 124 participants).
- "Neurodivergent" is itself contested; some scholars (Devon Price) treat it as social identity, others as disability category. The Hamilton & Petty (2024) *Frontiers* corrigendum acknowledged earlier under-citation of Spaeth & Pearson on this point.
- Several enthusiastic accounts (e.g., Faulkner University newsletter, Goblin Tools reviews, vendor blog posts) are non-empirical and read as marketing-adjacent; the keynote should rely on the peer-reviewed material above.
- The Stratalys/Special Olympics study sampled K-12 attitudes, not higher ed; the parent/educator findings translate as parallel signal, not direct evidence about postsecondary IDD students.
- **Aphantasia-AI literature is still largely first-person and theoretical**, not empirical; the fMRI-decoding work is research-stage, not deployed.
- Tim Shriver's "three percent locked out" framing refers to IDD specifically (which is distinct from the broader 15-20% ND population).
- The HEPI Policy Note 61's "18%" figure refers to students *including* AI-generated text directly in their work; the report explicitly cautions against reading this as cheating.
- The "AI doesn't amplify intelligence, it amplifies rehearsal" thesis is Josh's; the literature supports it implicitly (scaffolding, drafting, iteration are the dominant use cases) but no scholar has explicitly framed it that way — Josh would be staking new conceptual ground.

Completion Table

Research Plan Item	Status
1. UDL frameworks + AI integration	✓ CAST v3.0; Mallary 2025; CCC/UBC workshops

Research Plan Item	Status
2. Faculty adoption/training	✓ AHEAD, Mallary, NTU framework
3. Student-facing tools (Goblin/Otter/Glean/Speechify/Read&Write/NotebookLM/ChatGPT)	✓
4. DSO adapting; ADA/504/OCR	✓ PLOS One 2025; National Deaf Center; Hunton; Teachers College policy
5. ADHD-specific	✓ Springer ADHDvance LearnAR; Hamzah 2026; Frontiers Sweden
6. Autism-specific	✓ Ma et al. 2026; Williams Purdue
7. Dyslexia	✓ Keelor 2023; Song & Mao 2024; Read&Write; Snap&Read
8. Aphantasia (Josh's hook)	✓ Robbins; DEV.to first-person; Kerson AI Solutions
9. Executive function	✓ Goblin Tools; Frontiers EF + GenAI Sweden 2024
10. Landmark, Beacon, CIP, Cornell, Curry, Marshall, Drexel	✓
11. Special Olympics Inclusion Academy	✓ Stratalys 2024; Shriver; Cook
12. Critical/ethical perspectives	✓ Williams; Price; Hamilton & Petty
13. AI bias against autistic writing	✓ Williams; Liang et al. <i>Patterns</i>
14. Plagiarism detection bias	✓ Liang et al.; Bloomberg; AI Detector Pro; Kaltman
15. Recent 2023–2026 journal articles	✓ ten+ peer-reviewed cited

Research Plan Item	Status
16. Prevalence stats	✓ Doyle; NCES; YouGov; Purdue institutional
17. International (UK/Aus/Canada)	✓ Jisc; HEQCO; Loble; NTU
18. 2e/gifted ND	△ Touched via Price (gifted- as-masked-autism point); limited HE-specific empirical literature surfaced
19. Faculty/staff with ND using AI	✓ Williams Purdue; Inside Higher Ed Nov 2025
20. ND student voices/qualitative	✓ Atcheson CHI 2025; Pierrès; Zhao